



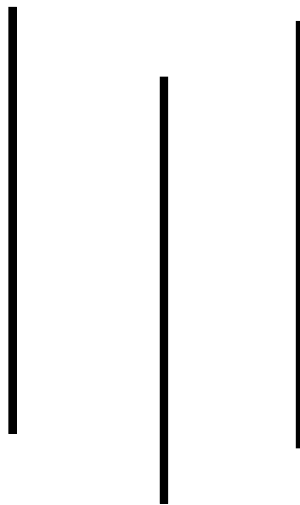
## Multiple View Geometry

### ***Lab2 – Feature Extraction and Image Registration***

Lab Sessions: 3 (6 hours)

Total Work Load: 15 hours

Programming Language: MATLAB



#### **Submitted by**

1. **Pravin Oli** (u1999699) [[pravin.oli.08@gmail.com](mailto:pravin.oli.08@gmail.com)]
2. **Gebrecherkos Gebrezgabhier** (u1999542) [[chereg2016@gmail.com](mailto:chereg2016@gmail.com)]

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**Note to the Professor:** We have decided not to include snip of MATLAB code in the report because it makes our report very long.

We request you to run following codes:

Lab2\_section2\_example.m  
Lab2\_section3\_step1.m  
Lab2\_section3\_step2.m  
Lab2\_section4\_step2.m  
Lab2\_section4\_step3.m  
Lab2\_section5\_RANSAC.m

## Section-1

### 1.Objectives:

- Gain familiarity with SIFT (Scale-Invariant Feature Transform) and planar transformations for rigid image registration.
- Learn to extract invariant features and their descriptors for image analysis.
- Understand feature matching techniques and their application in computing homography.
- Assess strengths and weaknesses of local feature-based approaches in image processing.

### 2.Methodology:

#### Section-2 (Image visualization and Shift features)

##### **Implementation:**

At first the two images (img101311.jpg and img101396.jpg) of a underwater coral reef patch in south of Florida, is loaded in MATLAB workspace using the script `lab2_section_2 example.m` file provided.

It uses `siftprecomputed` to load each image, extract SIFT descriptors, and retrieve keypoint locations. and `showkeys` to visualize the detected SIFT keypoints on each image.

**Output:**

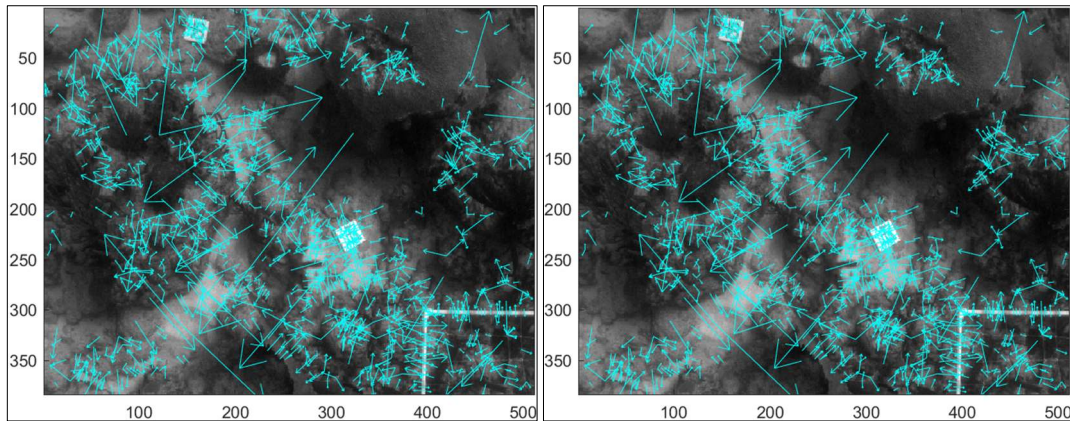


Figure1: *imgl01311.jpg* and *imgl01396.jpg* with shift key points and arrows

**Question:**

1. What is the meaning of the arrow's directions and lengths?

➔ The arrow direction shows the orientation of the keypoint. This is gradient dominant direction in the local area of the keypoint. The arrow length indicates the scale of the keypoint, which corresponds to the size of the detected feature. A longer arrow represents a larger scale (larger feature), while a shorter arrow represents a smaller scale.

### Section-3 (Feature Association)

Feature association involves matching feature descriptors between two images by measuring descriptor similarity, usually through distance calculation, to identify corresponding features. Location of corresponding feature pair is used to compute homography matrix which describe the transformation between two images.

**Implementation:**

Step1: In this step we created a script file `Lab2_section3_step1_match_sift_test.m` which call the function `matchsiftmodif.m` for extracting and associating SIFT features. It then returns two list of coordinates (CL1uv and CL2uv) with the locations of the associated features.

We tested this for images *imgl01311.jpg* (as image1) and *imgl01396.jpg* (as image 2), and values of distance ratio of 0.4, 0.6 and 0.8.

**Output:**

- Figure2 (Few Matches): This image likely corresponds to a `distRatio` of 0.4. The threshold is stricter, meaning only very reliable matches are retained, resulting in fewer feature associations.
- Figure3 (Moderate Matches): This image likely corresponds to a `distRatio` of 0.6. The threshold is more relaxed, allowing for additional matches, but still filtering out some ambiguous matches.

- Figure4 (Many Matches): This image likely corresponds to a distRatio of 0.8. With a higher threshold, more matches are considered valid, even if they are less distinct, resulting in a denser set of matches.

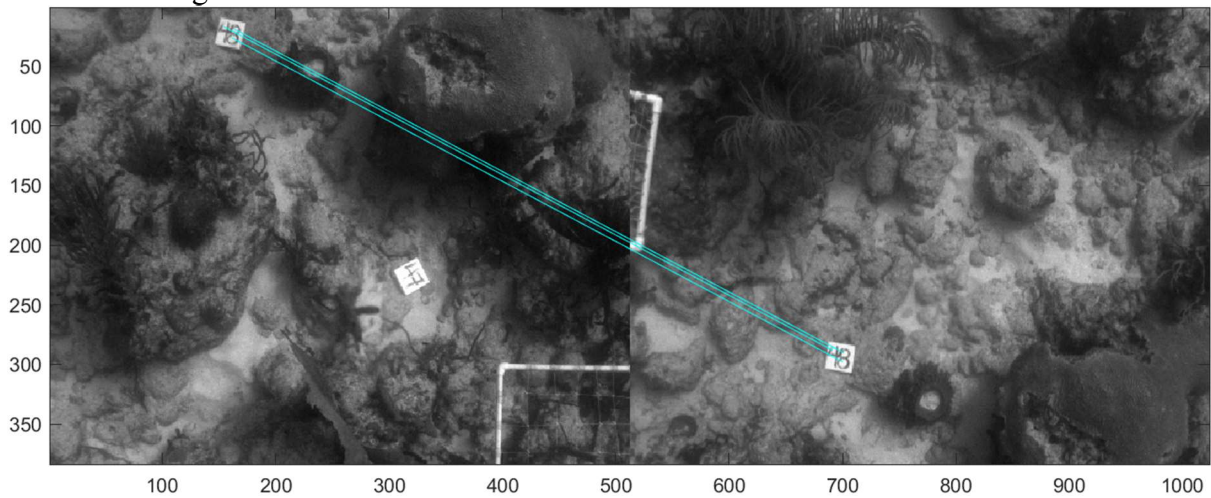


Figure2: Feature association for Dist. Ratio 0.4

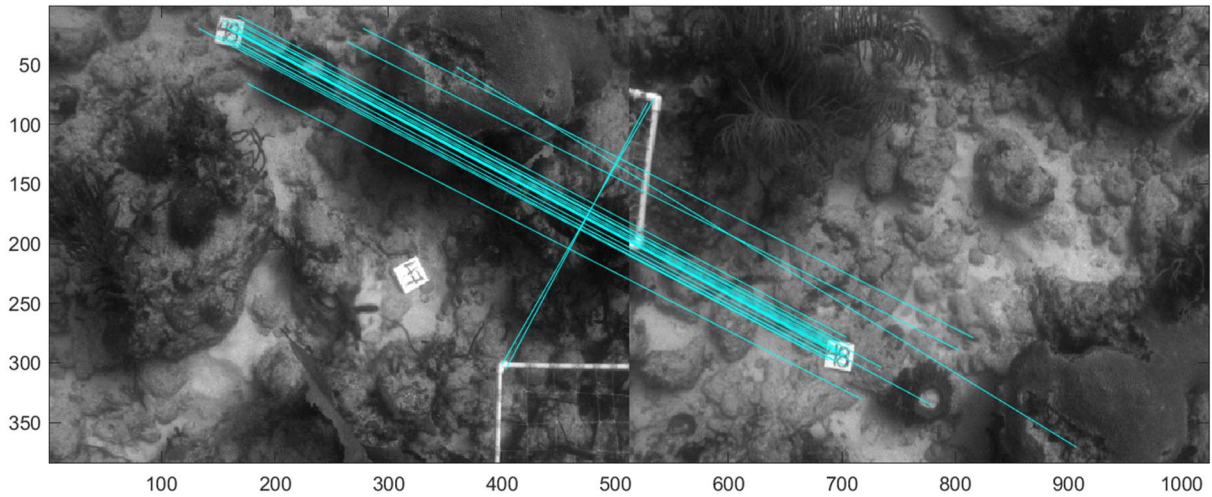


Figure3: Feature association for Dist. Ratio 0.6

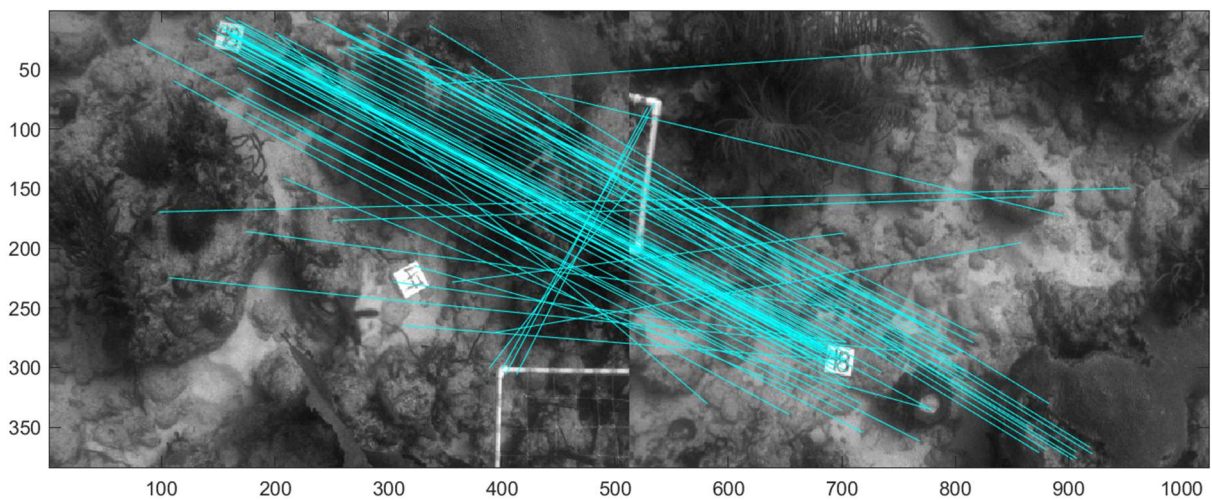


Figure4: Feature association for Dist. Ratio 0.8

**Question:**

1 What is the effect of the parameter distRatio in terms of number of associations?

➔The distRatio parameter has a direct impact on the number of associations (matches) between features in two images:

- Low distRatio value makes the matching process very strict. there are fewer matches overall, but they tend to be more reliable.
- High distRatio value allows more matches to pass the ratio test by relaxing the strictness slightly. This results in a high number of associations, but they may include some incorrect matches.

2. How can you tell visually that an association is wrong or not?

➔Correct matches often form same clusters or patterns of lines that follow the geometry of the scene, especially if the scene is largely planar. Incorrect matches may appear randomly scattered or isolated from the main pattern of matches. Also, we can look features, such as corners, edges, or textures that appear similar in both images.

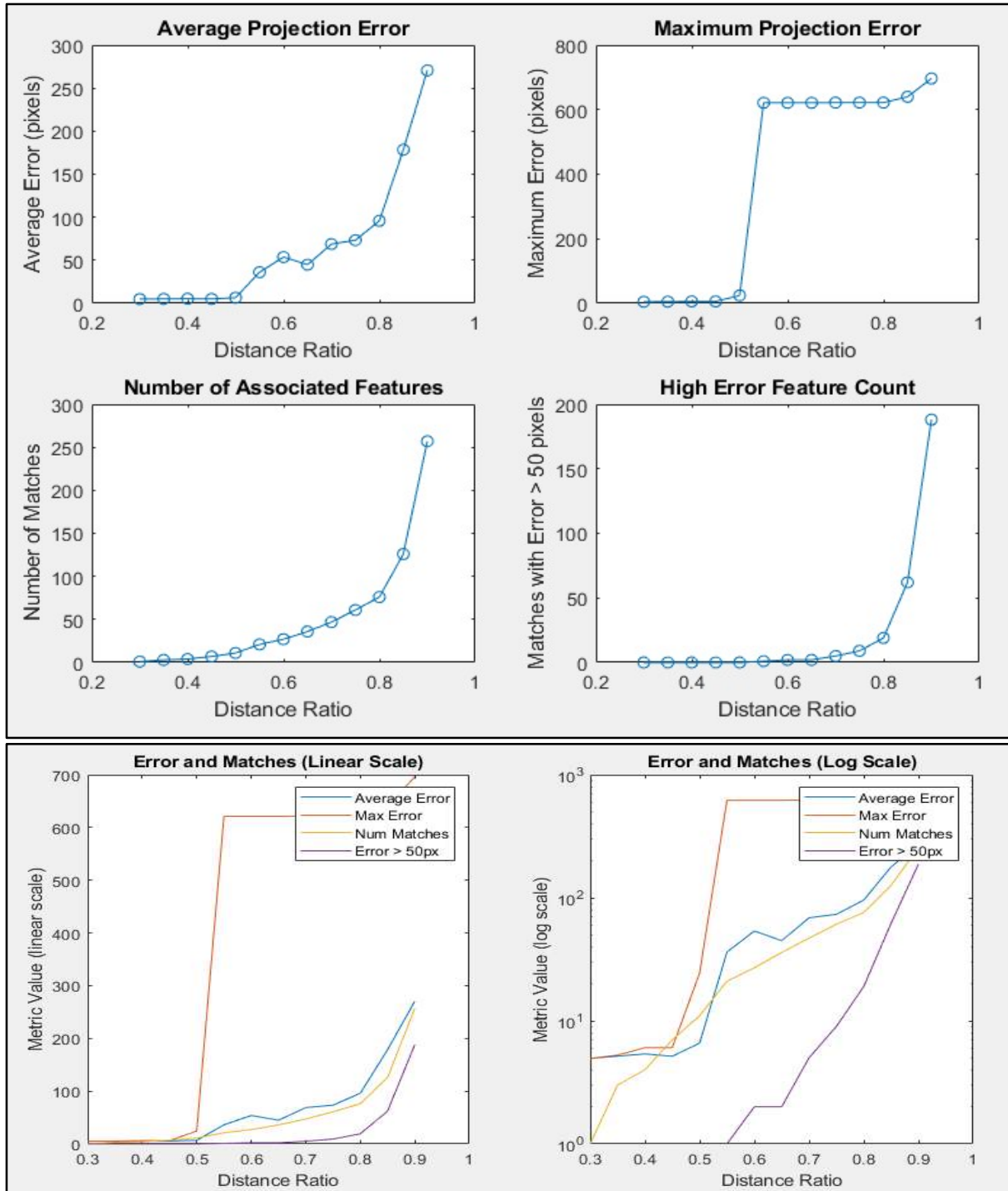
3. What is the effect of this parameter in terms of wrong associations?

➔For large distance ratio value, the more wrong associations we get. For small distance ratio, we get few associations but correct.

Step2: In this step we created a script file `Lab2_section3_step2.m` which call the function `projectionerrorvec.m` for returning an error vector that contains the Euclidean distance between each point in CL1uv and its corresponding point in CL2uv after applying the homography H12.

**Output:**

The error vector for increasing values of the distance ratio, from 0.3 to .09 by step of 0.05 is calculated and plotted in graph as below.



### Question:

1. What conclusion can we draw from this plot?
  - Is it possible to select a suitable value for the distance ratio for these images?
  - Is it possible to select value for the distance ratio that guarantees no outliers, while providing a high percentage of associations?

➔ From the above plot we can conclude that when distance ration increases, Average projection error, Maximum projection error, Number of associated features and High error feature count increases.

→ Yes, it is possible to select a suitable value for the distance ratio for these images. 0.5 value will be the best trade-off

→ For associations only I say yes but for correct associations it is No. because after 0.6 there is a sharp rise in the value of high error feature content.

## Section-4 (Estimating a Homography to Register a Pair of Images)

In this section, we first implemented a function to compute the homography matrix that defines planar transformation using a script file `computeHomography.m`. After this we proceed to step 2 and step 3 for this section.

### Implementation:

Step 2: In this step we created a script file `Lab2_section4_step2.m` which calls the function `showwarpedimages(I1, I2, H12)` for identifying the motion model that better fits the transformation of each image pair.

### Question:

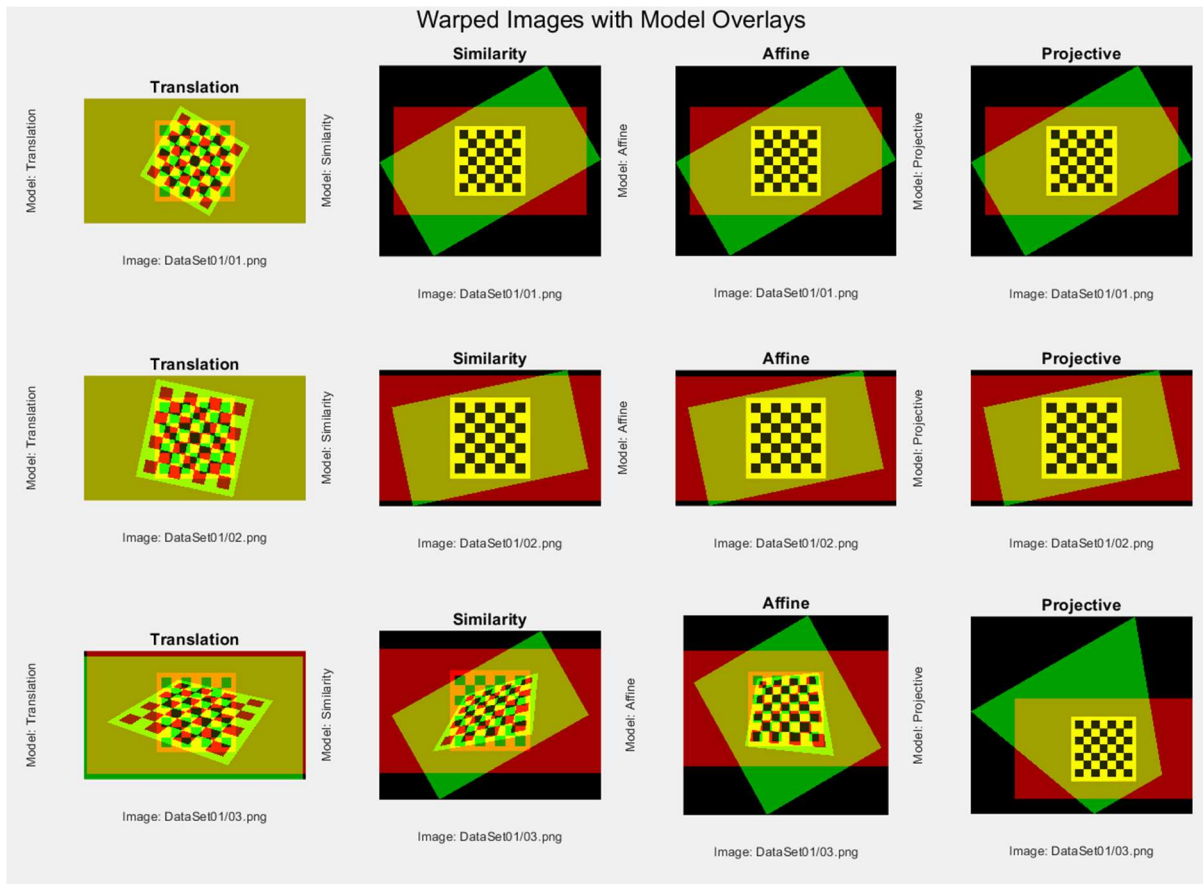
1. What are the motion models that better fit the planar transformation between the reference image and the rest?

→ Similarity and Projective for DataSet01/01.png but only Projective for DataSet01/02.png and DataSet01/03.png

### Output:

| Image sets       | Translation | Similarity | Affine  | Projective |
|------------------|-------------|------------|---------|------------|
| DataSet01/01.png | 138.0207    | 0.2319     | 0.2321  | 0.2319     |
| DataSet01/02.png | 88.0937     | 0.1595     | 0.1600  | 0.1569     |
| DataSet01/03.png | 170.0612    | 86.0312    | 27.9692 | 0.2542     |

- DataSet01/01.png: The Similarity and Projective models have the lowest error (0.2319), with the Affine model very close (0.2321). Translation has a significantly higher error, indicating it's not suitable. Similarity, Affine, and Projective all perform well, with Projective being marginally better for slight perspective alignment.
- DataSet01/02.png: The Projective model has the lowest error (0.1569), followed by Similarity (0.1595) and Affine (0.1600). Translation again shows high error, indicating it's not suitable.
- DataSet01/03.png: Projective model again has the lowest error (0.2542), while the Affine model (27.9692) performs much better than Translation and Similarity.



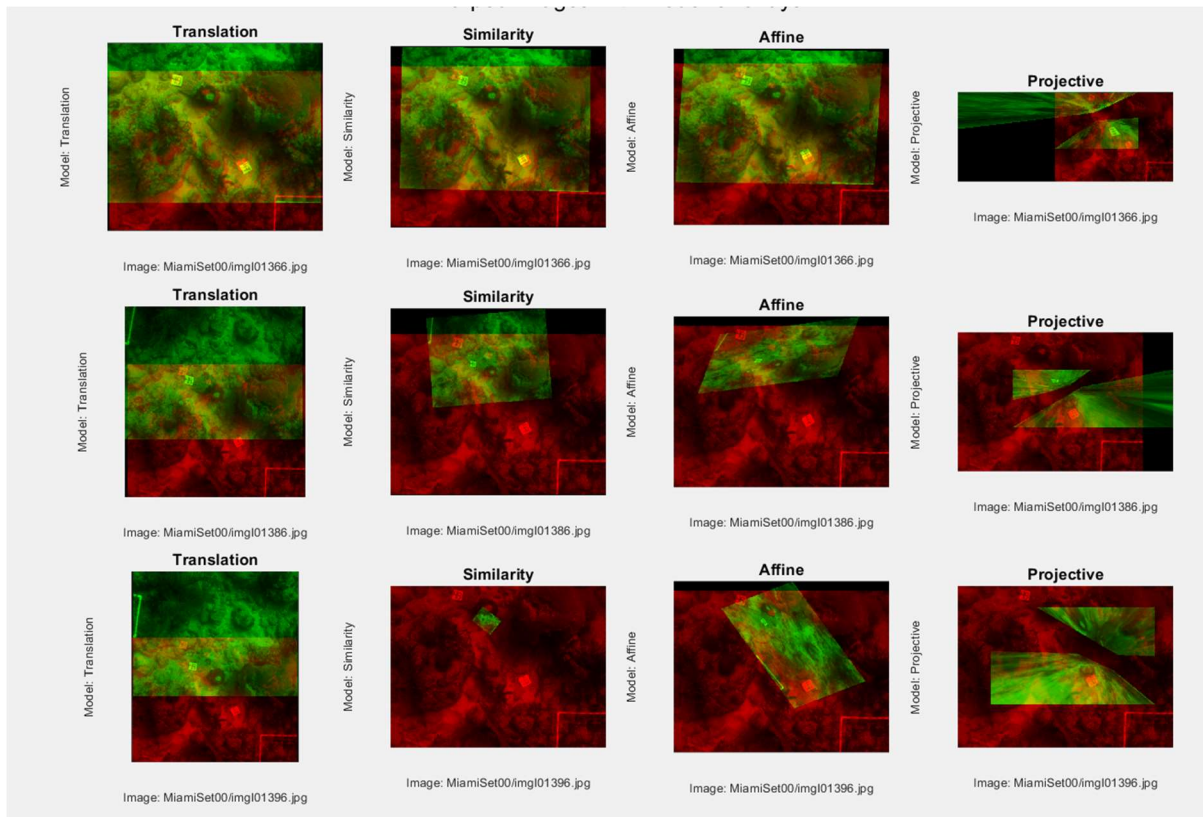
Step3: In this step we created a script file `Lab2_section4_step3.m` which call the function `matchsiftmodif.m` for computing the corresponding homography and displaying the results using overlay.

### Question:

- Does the obtained homography matrices allow registering the image pairs accurately, or are there significant misalignments?
- ➔ No, the obtained homography matrices allow registering the image pairs accurately. There are still significant misalignments.

### Output:

| Image sets               | Translation | Similarity | Affine  | Projective |
|--------------------------|-------------|------------|---------|------------|
| MiamiSet00/imgl01311.jpg | 0           | 0          | 0       | 0          |
| MiamiSet00/imgl01366.jpg | 31.6577     | 34.3136    | 35.0539 | 210.7689   |
| MiamiSet00/imgl01386.jpg | 86.6465     | 93.9925    | 93.2198 | 245.9491   |
| MiamiSet00/imgl01396.jpg | 132.7924    | 113.1932   | 95.9020 | 151.5267   |



### Section-5 (Improving the Registration Accuracy\_RANSAC)

This approach filters out outliers, allowing for a final homography computation based solely on reliable matches. As a result, RANSAC enhances alignment accuracy by reducing misalignments and distortions, making it a robust tool for image registration in variable and noisy conditions.

#### Implementation:

Step1: Initially we improved the homography function adding Ransac outlier rejection to the pipeline by creating `computeHomographyRANSAC.m` this step we created a script file.

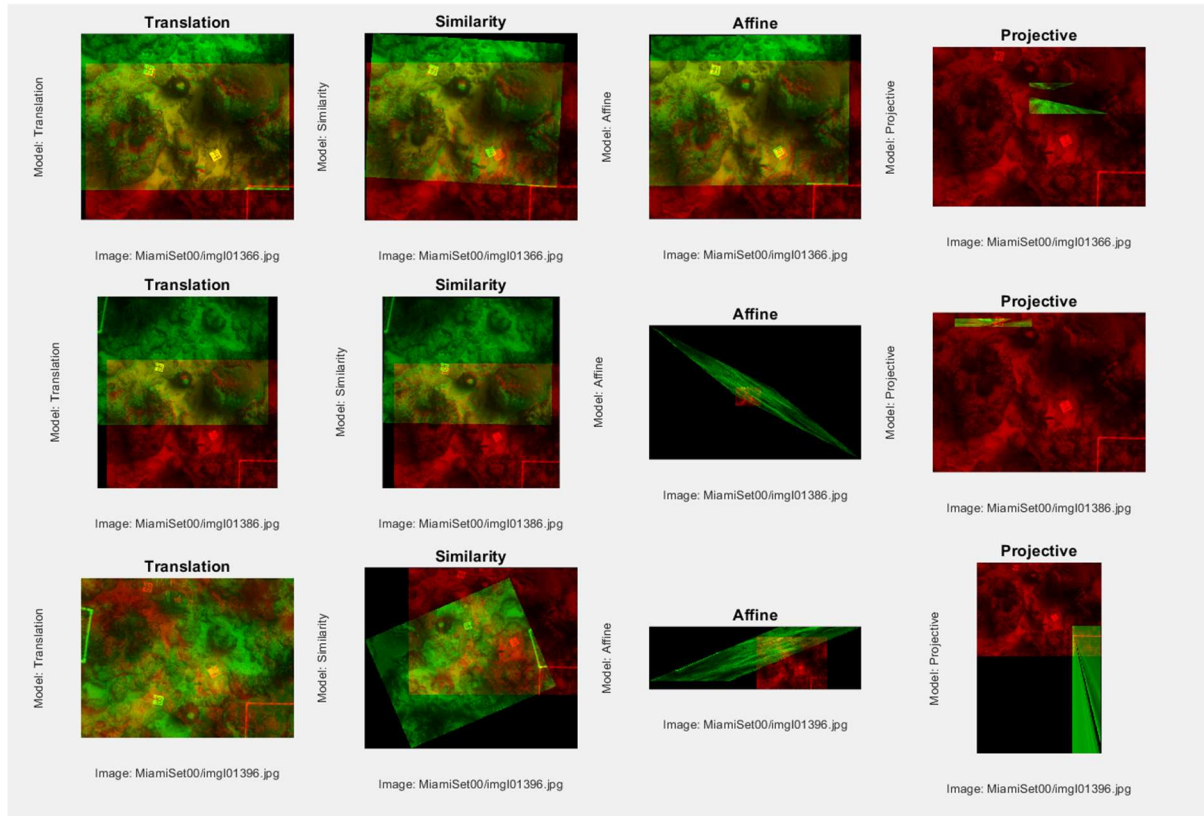
Step2: After that we created `Lab2_section5_RANSAC.m` which calls `computeHomographyRANSAC.m` for computing the homography matrices for the previously tested image pairs.

#### Question:

1. Are the results obtained with the RANSAC-based homography estimation more accurate than the previously obtained? The use of data normalization helps reducing the reprojection error?
- ➔ Yes, the results obtained with RANSAC-based homography estimation are generally more accurate than those obtained without it. By identifying and removing outliers, RANSAC ensures that only reliable matches (inliers) are used in the homography computation, reducing the influence of incorrect matches and thus lowering the overall reprojection error.

Normalization helps reduce the reprojection error, as it minimizes the effect of differences in scale, translation, and numerical precision, ensuring that the homography matrix accurately reflects the transformation between the images.

### Output:



### 3. Conclusion

In summary, these lab sessions introduced us to SIFT and planar transformations for aligning images. We learned how to extract and match features to compute homography matrices, enabling image registration. We also explored the strengths and limitations of using local features, understanding the importance of handling outliers with methods like RANSAC for more accurate results. These sessions provided a solid foundation in feature-based image processing techniques.